Apprehension

User Manual · Designed by Reyaan. · version 1.3.1 · Windows and macOS

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1. Quick start

1. **Put it on a sample.** In FL Studio, load Apprehension in an effect slot on the mixer track that holds your sample (open an empty slot's menu and pick Apprehension from the plugin list; run *Options > Manage plugins > Find installed plugins* first if you do not see it). Works the same in FL Studio for Windows and for Mac.
2. **Pick a preset.** Click the preset name at the top (or the arrows) and choose one of the 20 sample-flipping presets. Press play. That is all you need to get a sound.
3. **Adjust to taste.** Hover over any control for a one-line explanation. Turn **Mix** to set how much reversed sound you hear, and **Overall Pitch** to retune the whole sample.

> **Handy things to know:** double-click a knob to reset it · click the number under a knob to type a value · drag a corner of the window to resize it · hover over anything to see what it does.

2. How it works

Apprehension constantly records the sound that goes into it. Every so often it plays the last stretch of that recording **backwards**. Each stretch is called a **chunk**, and the **Division** (or **Time**) knob sets how long it is. Your original sound still passes straight through, and the **Mix** knob blends the two.

The reversed sound then travels through this chain, in order:

1. **Pitch (Reverse Pitch), Attack, Cutoff/Release, Smooth** - shape each chunk.
2. **Diffusion** - softens the sound.
3. **Low Cut and High Cut filters** - the point where **Feedback** is taken from, so repeats are filtered too.
4. **Chorus**, then **Flangus**, then **Reverb**.
5. **Ducking**, then **Width**.

After that it is mixed with your original sound (**Mix**), set to the **Output** level, and optionally run through **Signature Mode**'s auto-chop, then **Overall Pitch**, then **Soft Clip**. Everything in this last group affects *everything* you hear, not only the reversed layer.

> **Latency:** Overall Pitch adds a fixed 25 ms delay (always, so nothing shifts when you move the knob). The plugin tells your DAW, and FL Studio compensates automatically, so everything stays in time.

3. The window, control by control

The header

- **Logo and name** - Apprehension, designed by Reyaan.
- **Preset bar** - the arrows step through presets; click the name for the full list (Init sets everything to the default sound). After you move a knob the name shows "(edited)".
- **@Reyaan. Signature Mode** and **Freeze** buttons (explained below).
- **Theme** - change the look (section 6). **Manual** - opens this manual in your browser.

Time - the reverse engine

Control	Range / options	Default	What it does
Sync	On / Off	On	Chooses how the length of each reversed chunk is set. On = a musical note length (Division) that follows your project's tempo. Off = a fixed time in milliseconds (Time). If your host does not send a tempo, Sync falls back to Time.
Time Align	On / Off	Off	Without Time Align, the reversed chunks start whenever you press play and slowly drift against the beat (each chunk overlaps the next a little). With Time Align on, every chunk starts exactly on a grid line of your project, and the reversed sound mirrors around each beat: something that was 100 ms before a beat comes back 100 ms after it, and a hit that was exactly on a beat comes back exactly on a beat. It needs the project to be playing (it uses the host's beat position). Keep Smooth low if you want hits to stay sharp.
Division	1/16, 1/8T, 1/8, 1/4T, 1/4, 1/2, 1 bar, 2 bars	1/4	The note length of each reversed chunk when Sync is on. Short values (1/16, 1/8) give rhythmic, stuttery reverse hits. 1/4 and 1/2 are the classic reverse-into-the-beat sound. 1 bar and 2 bars give long, dramatic swells. T means triplet.
Time	10 to 4000 ms	500 ms	The length of each reversed chunk when Sync is off. Very short times (10-60 ms) turn into a gritty, pitched buzz; hundreds of milliseconds give classic reverse echoes.

Attack	0 to 2000 ms	0 ms	How long each reversed chunk takes to fade in. 0 ms starts at full volume. 100-400 ms gives a smooth swell. It is this knob that decides how fast the reversed sound starts.
Cutoff	5 to 100 %	100 %	Controls where each chunk stops. At 100% the whole chunk plays. At 50% only the first half of every chunk plays and then there is silence until the next chunk. Use it to make tight, gated reverse hits.
Release	1 to 2000 ms	30 ms	The fade-out time at the Cutoff point. Short = a clean chop. Long = the chunk tails away gently. You only hear it when Cutoff is below 100%.
Smooth	0 to 100 %	50 %	How much one reversed chunk overlaps and crossfades into the next. Low values keep every chunk sharp and punchy. High values blend the chunks into a continuous wash.

Attack, Cutoff and Release together make the shape of every reversed chunk: Attack = how fast it starts, Cutoff = where it ends, Release = how softly it ends.

Pitch

Control	Range / options	Default	What it does
Overall Pitch	-12 to 12 st	0 st	Shifts the pitch of everything the plugin outputs - both your original sound and the reversed layer - without changing the speed or timing. Use it to retune a sample to your key (the Key Detector tells you what key it will land in). Adds a fixed 25 ms of latency that your DAW compensates automatically. Quality is best within about 5 semitones; the further you go, the more the sound takes on a granular, robotic character.
Reverse Pitch	-24 to 24 st	0 st	Shifts only the reversed layer, like speeding up or slowing down a tape (so the chunk is read faster or slower). Combine it with Overall Pitch: the reversed layer's total shift is the two added together. Very high values shorten the longest chunk the plugin can play, because it needs to read further back in time.
Ref Hz	20 to 2000 Hz	440 Hz	A reference frequency for the pitch readout. Type a frequency (for example the pitch of the note you care about) and the panel shows what it becomes after the Overall and Reverse pitch changes, in Hz and as a note name, plus the difference in Hz. It does not change the sound.

Key detector (the panel under Pitch). It listens to your incoming sound for a few seconds, works out which musical key it is in (for example "A minor") and shows a bar for each of the 12 notes. Underneath it shows what that key becomes after your **Overall Pitch** change (and, in small writing, after the Reverse Pitch change too). A fraction of a semitone is shown in cents. It works best on melodic samples with a few seconds of music; drums, noise and very quiet sounds will confuse it, and it shows a confidence percentage so you know how sure it is. It also shows the pitch readout in Hz: the reference frequency you typed, and what it turns into.

Space & Modulation

Control	Range / options	Default	What it does
Feedback	0 to 95 %	25 %	Sends the reversed sound back into the buffer so it gets reversed again and echoes. 20-40% is a subtle trail, 60-80% builds long swirling repeats. The feedback is soft-limited, so it cannot run away.
Diffusion	0 to 100 %	20 %	Runs the reversed sound through a chain of all-pass filters that smear transients without changing the tone. Higher values soften hard hits into a gentle haze.
Width	0 to 200 %	100 %	Narrows or widens the stereo image of the reversed sound only. 0% collapses it to mono, 100% leaves it as it is, 200% pushes it extra wide.
Chorus	0 to 100 %	0 %	Thickens the reversed sound with a chorus. The Chorus Type knob picks the character; this knob sets how much.
Chorus Type	Classic, Ensemble, Dimension, Vibrato, Tape	Classic	Picks the kind of chorus. Classic = two voices, the familiar chorus. Ensemble = three slow voices, lush and wide. Dimension = four short voices with almost no obvious movement, just width. Vibrato = pitch wobble only, with no dry sound added. Tape = slow wow plus fast flutter, like an old tape machine.
Flangus	0 to 100 %	0 %	A sweeping comb-filter effect on the reversed sound, in the spirit of FL Studio's Fruity Flangus. The Flangus Type knob picks the character; this knob sets how much.
Flangus Type	Fruity, Jet, Metallic, Soft, Wide	Fruity	Picks the kind of flange. Fruity = the classic medium sweep. Jet = deep, slow and strong, like a passing plane. Metallic = negative feedback, a hollow ringing tone. Soft = gentle and subtle. Wide = the left and right sides sweep in opposite directions for a huge stereo sweep.

Chorus types: Classic, Ensemble, Dimension, Vibrato, Tape. **Flangus types:** Fruity, Jet, Metallic, Soft, Wide. They are described in the table above.

Ducking

Control	Range / options	Default	What it does
Amount	0 to 100 %	0 %	Turns the reversed sound down so it gets out of the way. 0% = no ducking. 100% = it disappears completely when triggered. What triggers it is set by the Pump button.
Release	10 to 1000 ms	200 ms	The recovery time after each duck. Short = snappy. Long = a slow swell back up. In Pump mode it is limited to one cycle of the chosen division.
Division	1/16, 1/8, 1/4, 1/2, 1/1	1/4	In Pump mode, how often the ducking happens: every 1/16, 1/8, 1/4, 1/2 or 1/1 (a whole bar). 1/4 is a pulse on every beat, like sidechain compression from a kick. It follows the host tempo.
Pump	On / Off	Off	Off = Input ducking: the reversed sound gets quieter whenever your

	original sound is loud, so the reversed layer fills the gaps. On = Pump: the reversed sound dips on every beat of the chosen Division, whatever you play - the classic sidechain pump.

Reverb

Control	Range / options	Default	What it does
Type	Room, Hall, Plate, Chamber, Cathedral	Hall	Picks the reverb's space. Room = small and quick. Hall = big and smooth. Plate = dense and bright, like a studio plate. Chamber = medium and warm. Cathedral = huge and slow. Size scales each one.
Mix	0 to 100 %	0 %	The amount of reverb. It is applied to the reversed sound, so the main Mix must be above 0% to hear it. The reverb keeps ringing after the reversed sound stops.
Size	0 to 100 %	50 %	Scales the size of the reverb space. Small = tight and close. Large = spacious and slow. Moving it while sound plays gives a gentle pitch glide in the tail.
Decay	0.2 to 15 s	2.5 s	The length of the reverb tail (the time it takes to fade by 60 dB). Under 1 s is tight. 2-4 s is natural. 8 s and up is an ambient wash.
Pre-Delay	0 to 250 ms	20 ms	A pause before the reverb begins. 20-40 ms keeps the front of the reversed sound clear before the reverb blooms; long pre-delays make a distinct echo-then-wash effect.
Damping	0 to 100 %	40 %	How much the reverb tail darkens as it fades. 0% keeps it bright and glassy. High values make it warm and dull, like a soft room.
Width	0 to 200 %	100 %	The stereo width of the reverb only.
Modulation	0 to 100 %	20 %	Gently modulates the reverb to avoid a metallic ring and add shimmer. Low values are natural; high values make the tail wobble like a chorus.

Filter

Control	Range / options	Default	What it does
Low Cut	20 to 2000 Hz (20 = off)	Off	A high-pass filter that removes low frequencies from the reversed sound. At its minimum (20 Hz) it is switched off. Use it to keep the reversed layer from muddying the bass.
Low Shape	6 dB, 12 dB, 18 dB, 24 dB, 36 dB, 48 dB, 72 dB, 96 dB, Brickwall	24 dB	The steepness of the Low Cut. 6 dB per octave is a very gentle tilt. 12 and 24 dB are the usual choices. 36 to 96 dB are very steep, and Brickwall is an almost vertical wall (32nd order) that removes everything below the frequency. Watch the graph - it draws the exact curve.
Low Reso	0 to 12 dB	0 dB	Boosts the frequencies right at the Low Cut corner for a more pronounced, vocal 'filter sweep' sound. 0 dB is neutral. It has no effect on the 6 dB shape.

High Cut	70 to 20000 Hz (20000 = off)	Off	A low-pass filter that removes high frequencies from the reversed sound. At its maximum (20 kHz) it is switched off. It goes all the way down to 70 Hz, so you can turn the reversed layer into a muffled low rumble.
High Shape	6 dB, 12 dB, 18 dB, 24 dB, 36 dB, 48 dB, 72 dB, 96 dB, Brickwall	24 dB	The steepness of the High Cut, from a gentle 6 dB per octave to a near-vertical Brickwall. The graph shows the exact curve.
High Reso	0 to 12 dB	0 dB	Boosts the frequencies right at the High Cut corner for a sharper, more 'synthy' filter sound. 0 dB is neutral. It has no effect on the 6 dB shape.

The graph shows the real curve made by both filters, from 20 Hz to 20 kHz. **Low Cut** removes everything below its frequency; **High Cut** removes everything above. Shapes run 6, 12, 18, 24, 36, 48, 72, 96 dB per octave and Brickwall. The filters act on the reversed sound only (your original sound is not filtered).

Output

Control	Range / options	Default	What it does
Mix	0 to 100 %	50 %	Blends your original (dry) sound with the reversed (wet) sound. It is an equal-power crossfade, so the loudness stays steady as you turn it. At 100% you hear only the reversed sound - turn on Time Align so it stays on the beat. The reversed sound is what the reverb, chorus, flangus, filters, ducking and width act on, so at 0% those are not heard.
Output	-24 to 12 dB	0 dB	The overall level leaving the plugin, from -24 dB to +12 dB. Use it to match your sample's loudness after adding effects.
Soft Clip	On / Off	Off	A gentle limiter-style safety. Anything below about -3 dB passes untouched; louder peaks are rounded off smoothly so the output can never go past 0 dBFS. Handy after adding feedback, chop or pitch effects.

The level meter under the knobs shows the peak output level in dB.

Header

Control	Range / options	Default	What it does
Freeze	On / Off	Off	Freeze stops the plugin from recording new audio and keeps playing the last chunk backwards in a loop. Press it at a moment you like to hold a sound; press it again to carry on. Your original sound keeps playing through the Mix as normal.
@Reyaan. Signature Mode	On / Off	Off	Switches on auto-chop. On every step of the grid the plugin does one thing to your sound: plays it

normally, silences it (gate), repeats a slice (stutter), plays it at half speed (half-time), slows it to a stop (tape-stop) or plays the last step backwards (reverse). It follows the host tempo (120 BPM if there is none). The window turns hot pink on black, with pink flashes on chop hits.

Signature Chop

Control	Range / options	Default	What it does
Intensity	0 to 100 %	100 %	The chance that each step actually does its action. At 100% the pattern plays exactly as shown. Lower values leave more steps untouched, for a looser, subtler chop.
Chop Mix	0 to 100 %	100 %	Mixes the chopped result with the normal sound. 100% = only the chopped sound. 50% = half and half, which makes the chops feel like accents on top of the groove.
Seed	0 to 99	0	A number from 0 to 99 that picks one 'version' of the randomness. The same seed always gives the same result, so you can find a variation you like and keep it. Roll seed picks a random one.
Rate	1/8, 1/16, 1/32	1/16	The length of each grid step. 1/8 = slow, big chops. 1/16 = standard. 1/32 = fast glitchy chops. A pattern is always 16 steps long.
Pattern	Auto, 1, 2, 3, 4, 5, 6, 7, 8, Custom	Auto	Picks the pattern of actions. Auto = a different built-in pattern each bar (the Seed changes which). 1-8 = one fixed built-in pattern. Custom = the steps you drew in the grid. Clicking a step in the grid switches this to Custom automatically.
Stutter Size	1/2, 1/4, 1/8	1/4	The size of the slice that repeats in a stutter step. 1/2 = two repeats per step, 1/4 = four, 1/8 = eight (fast, buzzy).
Gate	On / Off	On	Turn off to stop the chop from ever silencing a step. Steps that are switched off play normally.
Stutter	On / Off	On	Turn off to stop the chop from ever stuttering.
Half-time	On / Off	On	Turn off to stop the chop from ever slowing to half speed.
Tape-stop	On / Off	On	Turn off to stop the chop from ever doing a tape-stop.
Reverse	On / Off	On	Turn off to stop the chop from ever playing a step backwards.

4. Signature Mode (auto-chop)

Press **@Reyaan. Signature Mode** and the plugin starts auto-chopping your sound on the beat. The window also turns black and hot pink, with pink flashes on the chops (never more than about three a second).

The row of 16 boxes at the bottom is the pattern. Each box is one step, and each step does one of

these things:

Step	What happens
-	Plays normally.
gate	Silence for that step.
stutter	Repeats a short slice of the step (the size is set by Stutter Size).
half	Plays the step at half speed - half-time, one octave down.
tape	A tape-stop: it slows down to a stop within the step.
reverse	Plays the previous step backwards.

Make it your own:

- **Click a step** to change what it does (click again to move to the next kind; **right-click** goes backwards). The first time you click, Pattern switches to **Custom** and keeps whatever was showing, so you are editing from the current pattern. Your Custom pattern is saved with your project.
- **Clear steps** sets all 16 to "normal" so you can build a pattern from nothing.
- **Intensity** decides how many of the steps actually do their thing. **Chop Mix** blends the chopped result with the normal sound.
- **Rate** sets the step length (1/8, 1/16 or 1/32). **Stutter Size** sets how long a stutter slice is.
- **Pattern** chooses Auto, 1-8 or Custom. In Auto a different built-in pattern is picked every bar. **Seed** changes which patterns Auto picks and which steps Intensity skips; **Roll seed** picks a random seed. Same seed, same result.
- **Gate, Stutter, Half-time, Tape-stop, Reverse** switches let you ban any kind of step from happening at all.

The playing step lights up. The chop follows your project's tempo (it assumes 120 BPM if the host does not send one) and works on the final sound, so it chops both your original and the reversed layer together. Use **Time Align** too and everything stays locked to the beat.

5. The 20 presets

These are starting points for flipping samples. They are all built from the same knobs you see, so after loading one you can tweak anything. Load a preset, put it on your sample, then adjust **Mix**, **Overall Pitch** and, for the chopping ones, the **Seed**. Most presets use Time Align, so press play for them to lock to the beat.

#	Name	What it is for	Key settings
01	Reverse Swell	The classic riser: the sample plays backwards into every half-note beat, locked to the grid.	Mix 100 %, Division 1/2, Type Hall, Mix 18 %
02	Backwards Dream	A slow, washed-out reverse bloom across a whole bar - for pads, vocals and long samples.	Mix 75 %, Division 1 bar, Type Hall, Mix 40 %
			Mix 35 %, Division 1/4, @Reyaan.

03	**Half-Time Flip**	Drops parts of the sample into half-time on the 1/16 grid for that slowed, heavy bounce.	Signature Mode On, Pattern 5, Overall Pitch -2 st, Type Room, Mix 10 %
04	**Chopped & Screwed**	Pitched down and chopped with tape-stops and stutters - the slowed-down Houston flip.	Mix 45 %, Division 1/2, @Reyaan. Signature Mode On, Pattern Auto, Overall Pitch -4 st, Type Chamber, Mix 25 %
05	**Vocal Flip Up**	Raises a vocal sample up a few semitones with a light reverse shadow behind it.	Mix 30 %, Division 1/4, Overall Pitch +4 st, Type Plate, Mix 14 %
06	**Soul Chop**	Rearranges a soul loop with gentle stutters, reverse slices and gaps. Change the Seed for new variations.	Mix 20 %, Division 1/4, @Reyaan. Signature Mode On, Pattern Auto, Type Room, Mix 12 %
07	**Stutter Roll**	Fast 1/32 stutters and gates for drum-roll style rolls and glitchy fills.	Mix 15 %, @Reyaan. Signature Mode On, Pattern 8
08	**Tape Stop Drop**	Plays the sample straight, then a tape-stop slows it to a halt at the end of every bar.	Mix 10 %, @Reyaan. Signature Mode On, Pattern 7, Type Plate, Mix 10 %
09	**Reverse Echo**	A backwards echo that repeats and darkens, ducking out of the way while the sample plays.	Mix 60 %, Division 1/8, Type Room, Mix 12 %
10	**Pump Reverse**	Reversed sound that pumps on every quarter note, like sidechain compression. Great on pads.	Mix 70 %, Division 1/2, Type Plate, Mix 30 %
11	**Lo-Fi Flip**	Dusty, warm and slightly wobbly - tape chorus, rolled-off highs and soft clipping.	Mix 40 %, Division 1/4, Type Room, Mix 12 %
12	**Dusty Soul Loop**	A slightly lowered, warm loop with an occasional tape-stop. Quick to make a sample feel old.	Mix 25 %, Division 1/2, Overall Pitch -1 st, @Reyaan. Signature Mode On, Pattern 7, Type Chamber, Mix 18 %
13	**Boom Bap Reverse**	Tight, punchy reverse hits that stop early - sits nicely under 90s-style drums.	Mix 55 %, Division 1/2
14	**Ambient Wash**	A huge two-bar reverse bloom with a cathedral tail. Turns any sample into a pad.	Mix 85 %, Division 2 bars, Type Cathedral, Mix 60 %
15	**Vaporwave Slow**	Pitched down with dimension chorus and a soft flange for a hazy, slowed-down mood.	Mix 40 %, Division 1 bar, Overall Pitch -3 st, Type Hall, Mix 45 %
16	**Jet Reverse**	A reversed sound with a deep, slow jet-plane flange sweeping through it.	Mix 60 %, Division 1/4T, Type Plate, Mix 15 %
17	**Metal Rise**	A reverse swell pitched up an octave with a metallic flange - a bright riser for transitions.	Mix 70 %, Division 1/2, Reverse Pitch +12 st, Type Room, Mix 10 %
18	**Frost Pad**	A frozen-feeling pad of endless	Mix 70 %, Division 1 bar, Type

		reverse trails. Press Freeze on a moment you like to hold it.	Cathedral, Mix 65 %
19	**Glitch Flip**	Rapid 1/32 glitches with reversed layers underneath. Roll the Seed for endless variations.	Mix 100 %, Division 1/4, @Reyaan. Signature Mode On, Pattern Auto, Type Plate, Mix 20 %
20	**Octave Down Flip**	Drops the whole sample an octave for a heavy, bassy flip, with the highs tucked away.	Mix 25 %, Division 1/4, Overall Pitch -12 st

6. Themes

Only the black-and-pink look is built in. You (and anyone you share with) add the others yourselves. Click **Theme**:

- **Create a theme template** writes a file called *My Theme.json* into the themes folder and shows it to you. The folder is `%APPDATA%\Reyaan\Apprehension\Themes` on Windows and `~/Library/Application Support/Reyaan/Apprehension/Themes` on a Mac.
- Open the file in Notepad (or TextEdit on a Mac - choose Format > Make Plain Text), change the colours (written like `#ff3d9a`), save it, then choose **Reload themes**. Your theme is now in the Theme menu. Send the .json file to a friend and they can drop it into the same folder.
- **Open themes folder** opens the folder in Explorer (Windows) or Finder (Mac).

A theme can set the background gradient, glow colours, knob colours, panel colour and opacity, text colours, and a font name (any font installed on the computer). It can also point at a `"image"` (a still picture that slowly pans and zooms behind the plugin) or a `"frames"` folder of numbered pictures (for example exported from your own video with `ffmpeg -i clip.mp4 -vf fps=24,scale=640:-1 frames/%04d.jpg`) that plays as a loop. Only use pictures and footage you own or have permission to use.

The theme you choose is saved with your project.

7. Recipes for flipping samples

1. **Reverse riser into the drop:** load *Reverse Swell*; raise Attack for a longer swell; put Mix at 100%. Automate Mix or Cutoff to build tension.
2. **Instant new loop from a chord sample:** load *Soul Chop*, then try Seed values until the arrangement feels right. Use Chop Mix at 50% to keep the original groove under it.
3. **Re-key a sample to your beat:** look at the Key Detector, decide which key you want, and turn **Overall Pitch** until the "After" key matches. It does not change the speed, so it stays in time.
4. **Slowed and lo-fi:** load *Vaporwave Slow* or *Lo-Fi Flip* and lower Overall Pitch by 2-4 semitones. Use High Cut to darken it.
5. **Keep the groove clear:** turn on Ducking in Input mode so the reversed layer drops while the original plays, or use Pump with 1/4 to make it breathe with the kick.
6. **Hold a moment:** press **Freeze** on a nice moment; the last chunk loops backwards until you press it again. Add reverb and chorus for a pad.

8. Questions and troubleshooting

I hear nothing different. Check **Mix** is above 0%. The reverb, chorus, flangus, filters and ducking only act on the reversed sound, so they need some Mix. Also check Signature Mode and Time Align need the project to be playing.

Time Align does nothing / sounds late. It follows the host's beat position, so press play. If your DAW's delay compensation is switched off, the plugin's 25 ms of latency will not be corrected - turn it back on (FL Studio normally does this automatically unless you switched it off in its settings).

The sound is too quiet or too loud after a preset. Presets are level-matched approximately; use **Output** and, if needed, **Soft Clip**.

The reverb tail keeps going. That is the tail of the reverb (up to 15 s). Lower Decay or Mix.

The Key Detector says "listening...". It needs a few seconds of audio that is not too quiet. It reads your original (dry) sound, before any effect.

The text looks different on my computer. The plugin uses the first cursive font it finds on your computer (Apple Chancery or Snell Roundhand on a Mac; Monotype Corsiva, Palace Script, Edwardian Script, Brush Script, Segoe Script or Lucida Handwriting on a PC). A theme can name a different installed font.

Mac: macOS says the installer or plugin 'cannot be opened' or is 'from an unidentified developer'. This happens with any app or plugin that has not been signed with a paid Apple developer certificate. Right-click (or Control-click) the installer, choose **Open**, then **Open** again. If it was blocked already, open *System Settings* > *Privacy & Security*, scroll down and click **Open Anyway**. If your host still refuses to load the plugin, open Terminal and run `xattr -cr /Library/Audio/Plug-Ins/VST3/Apprehension.vst3` (and the same for `/Library/Audio/Plug-Ins/Components/Apprehension.component`), then rescan plugins.

Mac: where are the plugins installed? The VST3 goes to `/Library/Audio/Plug-Ins/VST3` and the Audio Unit (for Logic and GarageBand) to `/Library/Audio/Plug-Ins/Components`. The manual is also placed in `/Library/Application Support/Reyaan/Apprehension`. In FL Studio for Mac use *Options* > *Manage plugins* > *Find installed plugins*.

Pink flashing. Signature Mode flashes the background on some chop hits, at most about three times a second. If flashing bothers you, do not use Signature Mode, or ask for a calmer theme.

Where are my settings saved? Everything, including your Custom chop pattern, the preset name and the theme, is saved inside your project file.